Ames Housing Price Prediction

Project Summary Report

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1. Project Objective

The goal of this project was to build a reliable machine learning pipeline that predicts residential SalePrice for the Ames, Iowa housing dataset (2,930 records, 79 raw features from official assessor records), while following a rigorous, leakage-free workflow from data understanding through final model selection.

2. What Was Done — End-to-End Workflow

- Renamed all raw columns to clear, readable names and performed full dataset inspection (shape, types, duplicates, missing values).
- **Train / Validation / Test split** (64% / 16% / 20%) performed first, before any exploration, stratified on SalePrice quintiles to keep the split leakage-free from the start.
- **Exploratory Data Analysis (EDA):** distribution checks, outlier identification, correlation analysis, and 13 analytical questions answered on the training set only (e.g. neighborhood price patterns, effect of quality/bedrooms/bathrooms, seasonal trends).
- **Preprocessing pipeline:** domain-aware missing-value handling, ordinal and one-hot encoding, RobustScaler, and a log1p transform on the right-skewed SalePrice target, all built with scikit-learn ColumnTransformer/Pipeline and fit on training data only.
- **Feature selection:** variance threshold + correlation-redundancy filtering applied after encoding, reducing the encoded feature space from ~241 to 167 columns.
- **Reduced 26-feature set (bonus track):** a practical, self-reportable subset of the 79 raw features, described in full in Section 5 below.
- **Modeling:** SVR, Random Forest, and XGBoost were each tuned with cross-validated search (GridSearchCV for SVR, RandomizedSearchCV for RF/XGBoost), then combined into a Stacking ensemble (RF + XGBoost + SVR, Ridge meta-learner). The exact same tuned configuration was then re-applied — not re-searched — to the reduced 26-feature pipeline, to test how much accuracy that simpler input gives up.
- **Evaluation:** all 8 resulting models (4 algorithms × 2 feature sets) were scored on Train, Validation, and Test splits; the Test set was touched exactly once, after every model was already finalized.

3. Models Trained & Final Results (Test Set)

Tuned hyperparameters — Full 79-feature pipeline:

- SVR: kernel = rbf, C = 1, epsilon = 0.01, gamma = scale
- Random Forest: n_estimators = 300, max_depth = 16, min_samples_split = 5, min_samples_leaf = 1, max_features = 0.5
- XGBoost: n_estimators = 300, max_depth = 3, learning_rate = 0.1, subsample = 1.0, colsample_bytree = 0.7, reg_alpha = 0.5, reg_lambda = 1

Model	Feature Set	Test RMSE ($)	Test MAE ($)	Test R²
Stacking (RF+XGB+SVR)	Full (79→167)	22,442	12,715	0.927
SVR (tuned)	Full (79→167)	23,105	12,940	0.923
Stacking (RF+XGB+SVR)	Reduced (26→60)	23,872	14,941	0.918
XGBoost (tuned)	Full (79→167)	24,141	14,061	0.916
SVR (tuned)	Reduced (26→60)	24,556	15,568	0.913
Random Forest (tuned)	Full (79→167)	25,624	15,554	0.905
XGBoost (tuned)	Reduced (26→60)	26,121	16,070	0.902
Random Forest (tuned)	Reduced (26→60)	26,657	16,584	0.898

- **Best overall model:** Stacking ensemble (RF + XGBoost + SVR) on the full 79-feature pipeline — Test RMSE = $22,442, R² = 0.927. It narrowly beats standalone SVR and clearly outperforms single tree-based models.
- **Full vs. Reduced gap:** switching from 79 to 26 features costs only $1,000–$2,000 extra RMSE and about 0.008–0.014 R² across every algorithm — a small, acceptable trade-off for a much simpler user-facing input.

4. Data Leakage & Overfitting — What We Checked

Data leakage: prevented, not detected after the fact

- The train/validation/test split was performed in Section 1, before any EDA or preprocessing decision — the single biggest leakage guard in the project.
- Every preprocessing statistic (imputation medians, encoding categories, scaler parameters, feature-selection thresholds) was fit on training data only, then only applied (never re-fit) to validation and test.
- Every hyperparameter search (GridSearchCV / RandomizedSearchCV) used training data only inside its internal cross-validation folds; validation and test rows were never part of any search.
- The Stacking meta-learner was trained on out-of-fold base-model predictions (cv=5), so it never saw a prediction a base model made on a row it was trained on.
- The test set was used exactly once, after every model in both tracks (full and reduced) was already finalized.
- **Conclusion:** No data leakage was found anywhere in the pipeline.

Overfitting: mild, and concentrated in one model

Model	Train R^2	Test R^2	Train–Test Gap
Random Forest (tuned)	0.979	0.905	0.074 (largest)
XGBoost (tuned)	0.972	0.916	0.056
Stacking (RF+XGB+SVR)	0.973	0.927	0.045
SVR (tuned)	0.952	0.923	0.029 (smallest)

- **Random Forest overfits the most** (0.074 train–test gap) — expected, since deep trees can memorize training rows even after tuning max_depth/min_samples_leaf.
- **SVR generalizes best** (0.029 gap) — its limited capacity relative to the number of features keeps it from memorizing the training set.
- **Stacking sits in between** (0.045 gap) and still achieves the best Test performance overall, since its Ridge meta-learner is trained on out-of-fold predictions rather than in-sample ones.
- **No underfitting was observed:** all four models reach Test R^2 between 0.90 and 0.93, close to the ~0.89 in-sample linear R^2 ceiling estimated at the preprocessing stage — every model captured most of the available signal.

5. Extra Contribution: The Reduced 26-Feature Idea

Beyond the required 79-feature pipeline, the team explored a practical question: could an ordinary home seller — without a professional inspection — fill in a much shorter form and still get a usable price estimate?

Each of the 79 raw features was audited against one test: “Can a typical seller answer this without expert help?” Features needing technical judgment (e.g. BasementQuality, KitchenQuality), rarely-known construction details (e.g. RoofMaterial, ElectricalSystem), or information that only exists after a sale happens (e.g. SaleType, MonthSold — which would leak post-sale information into a pre-sale prediction) were excluded. This left 26 self-reportable features covering size, room counts, year built, location, and general condition.

Statistics were used only afterward, as a check — not as the selection method. The correlation review confirmed the choice was sound: the 26-feature set's average correlation with SalePrice (0.454) was actually higher than the full set's (0.326), and its in-sample linear R^2 (0.881) was within about 1% of the full set's (0.892).

- **Result:** cutting 67% of the raw input fields cost less than 1.2% of linear R^2, and about 1–2 cents on the dollar of Test RMSE once modeled with the same tuned hyperparameters — a strong case for offering a simplified estimator alongside the full assessor-based model.
